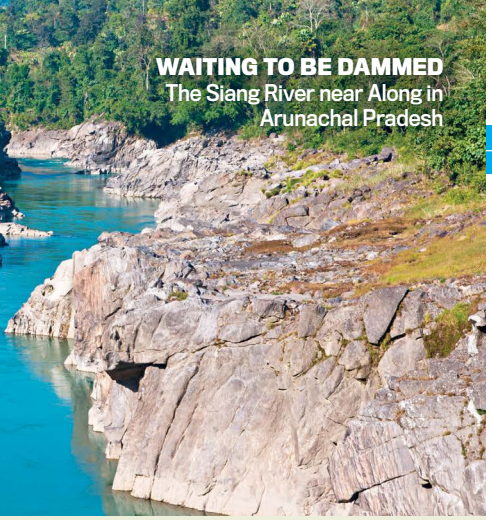


WAITING TO BE DAMMED
The Siang River near Along in Arunachal Pradesh

Upper Multipurpose Project in Arunachal Pradesh. Additionally, large reservoirs and artificial lakes should be developed to store excess water released by China, reducing the risk of sudden floods and water shortages

↘ A robust disaster management system needs to be established to effectively respond to unexpected water surges or supply disruptions caused by Chinese dam operations

impacts on India and Bangladesh, framing the project as a climate-friendly solution that will power 300 million people while curbing fossil fuel use. Backing this stance, Chinese foreign ministry spokesperson Guo Jiakun says the project underwent rigorous scientific evaluation and poses no risk to downstream ecosystems, geological stability or water rights. Instead, Beijing argues it will help mitigate floods and aid climate adaptation in India and Bangladesh. However, China's assurances on the Medog Dam ring hollow, given its track record of secrecy, unilateralism and broken commitments on transboundary rivers. Independent researcher Gabriel Lafitte has highlighted that past Chinese project announcements have been long on statistics but short on execution details.

India's Counter

A decade ago, India embarked on a defensive hydropower strategy to counter China's upstream dam-building spree on the Brahmaputra. Rooted in the international legal principle of "prior appropriation"—which grants water rights to the first user—India's plan aims to pre-emptively secure control over the river's flow. "India can still challenge Beijing by constructing its own dams near the border, keeping China on edge. This would give New Delhi some leverage in potential water negotiations with Beijing," says Kugelman.

After Beijing announced the Medog Dam, India has accelerated plans for the Siang Upper Multipurpose Project (SUMP)—a colossal hydropower dam in Arunachal Pradesh intended to counter China's influence over the river. With a proposed capacity of 11,000 MW, SUMP's reservoir would hold 9.2 billion cubic metres (bcm) of water, compared to Medog's 5.5 bcm. Estimated at $17 billion (Rs 1.5 lakh crore), it will be India's most powerful hydroelectric project, designed to regulate water flow, mitigate seasonal droughts and serve as a safeguard against sudden surges from Chinese dams. While feasibility surveys are underway in Parong, a remote hamlet in Siang district, SUMP has faced resistance since its proposal by NITI Aayog in 2017. The project threatens to submerge over 30 villages, putting thousands at risk of displacement.

Experts also warn against escalating the dam race, as it could prompt China to fast-track its own projects, capitalising on its superior execution speed to shift the balance further in its favour. Such a move might also strain relations with Bangladesh, a crucial downstream stakeholder in any future Brahmaputra basin management framework. "The concerns we've raised about China's projects are equally relevant to Bangladesh," notes Prof. B.R. Deepak of the Centre for Chinese & Southeast Asian Studies at JNU.



"New Delhi needs to start thinking now about how to mitigate the possible deleterious implications—especially for water security, the environment and geopolitics"

MICHAEL KUGELMAN
Director, South Asia Institute, Washington

Rather than pursuing a spree of dam construction, experts urge India to strengthen its northeastern water management systems. This includes enhancing independent river flow monitoring from China using advanced satellite technology, refining flood risk assessments and upgrading telemetry stations. Gamble points out that India lags behind China in studying Himalayan ecology and river systems. "India has yet to thoroughly analyse the flow data China has provided on the Yarlung Tsangpo. Instead of issuing threats over dam construction, India must invest in a deeper understanding of the region's hydrology and risks. Armed with this knowledge, it can engage China in informed negotiations, presenting concrete evidence on the downstream impacts," she asserts.

India and China share several major rivers, including the Brahmaputra, Sutlej and Indus. However, water-sharing remains a contentious issue, in the absence of a formal treaty. Despite China's upper riparian advantage, India can push for stronger data-sharing commitments under international conventions like the UN Watercourses Convention, which mandates that no country can take actions significantly harming another. However, neither India nor China is a signatory, and no Brahmaputra basin nation has ratified the 2014 UN Convention on Non-Navigational Water Uses, rendering first-user rights unenforceable.

In the Brahmaputra's turbulent waters, survival will depend not on the might of dams but on the foresight of nations. ■

- with India Today NE Bureau